

Dynamic Guideline: WHO-Based Meningitis - Diagnosis and/or Treatment and/or Management

I. WHO Latest Guideline Summary (Summary Date: 29 April 2026)

1.1 Guideline Overview

Field	Details
Guideline Title	WHO guidelines on meningitis diagnosis, treatment and care
Publication Date	2025 (latest update referenced 29 April 2026 via: "Meaningful youth engagement in a WHO guideline development process: case study of WHO meningitis guideline")
Target Population	Neonates, infants, children, adolescents, and adults with suspected or confirmed bacterial, viral, fungal, or tuberculous meningitis in all healthcare settings
Scope and Objectives	To provide evidence-based recommendations for the diagnosis, empirical and targeted treatment, supportive care, and follow-up management of meningitis across all resource settings

Related WHO Guidelines Reviewed (5 Guidelines)

#	Guideline Title	Year	Focus
1	WHO guidelines on meningitis diagnosis, treatment and care: executive summary	2025	Comprehensive diagnosis, treatment, and care
2	Defeating meningitis by 2030: a global road map	2021	Global strategy for prevention, diagnosis, treatment, and aftercare
3	WHO guidance on the use of meningococcal vaccines (Position Paper)	2024	Vaccination strategies for Neisseria meningitidis
4	WHO guidelines on management of Tuberculous Meningitis	2022 (updated)	Diagnosis and treatment of TB meningitis
5	Meaningful youth engagement in a WHO guideline development process: case study of WHO meningitis guideline	29 April 2026	Guideline development methodology incorporating youth engagement

1.2 Key Recommendations

A. Diagnosis

Recommendation	GRADE Evidence Quality	Strength
Lumbar puncture (LP) and cerebrospinal fluid (CSF) analysis should be performed in all patients with suspected meningitis unless contraindicated	High	Strong
CSF Gram stain, culture, and molecular testing (PCR) should be used for pathogen identification	High	Strong
Blood cultures should be obtained before antibiotic administration in all suspected bacterial meningitis cases	Moderate	Strong
Rapid diagnostic tests (latex agglutination, lateral flow assays) should be used where molecular testing is unavailable	Moderate	Conditional
Neuroimaging (CT/MRI) should be performed before LP only in patients with specific clinical contraindications (focal neurological signs, papilloedema, GCS $\leq$ 12, seizures)	Moderate	Strong
CSF opening pressure measurement should be performed during LP	Low	Conditional
Metagenomic next-generation sequencing (mNGS) may be considered for cases with negative conventional testing	Low	Conditional

B. Empirical Treatment

Recommendation	GRADE Evidence Quality	Strength
Empirical antibiotics should be administered within 1 hour of clinical suspicion of bacterial meningitis	High	Strong
Neonates (0–28 days): Ampicillin + gentamicin (or cefotaxime in settings with high gram-negative resistance)	Moderate	Strong
Infants and children (1 month–17 years): Third-generation cephalosporin (ceftriaxone or cefotaxime) $\pm$ vancomycin where penicillin-resistant pneumococcus is prevalent	High	Strong
Adults: Third-generation cephalosporin + ampicillin (for <i>Listeria</i> coverage in patients >50 years or immunocompromised) $\pm$ vancomycin	High	Strong
In resource-limited settings where cephalosporins are unavailable, high-dose parenteral ampicillin or penicillin + chloramphenicol is an acceptable alternative	Moderate	Conditional
Dexamethasone (0.15 mg/kg q6h for 4 days) should be administered before or with the first dose of antibiotics in suspected bacterial meningitis in high-income settings	High	Strong
Dexamethasone use in low-income settings: recommended for pneumococcal meningitis; evidence uncertain for other pathogens	Low	Conditional

C. Targeted Treatment

Recommendation	GRADE Evidence Quality	Strength
<i>Streptococcus pneumoniae</i> : Ceftriaxone (or cefotaxime) for 10–14 days; add vancomycin if MIC to penicillin > 2 µg/mL	High	Strong
<i>Neisseria meningitidis</i> : Ceftriaxone for 5–7 days (or penicillin G if susceptible)	High	Strong
<i>Listeria monocytogenes</i> : Ampicillin + gentamicin for 21 days	Moderate	Strong
<i>Haemophilus influenzae</i> type b: Ceftriaxone for 7–10 days	High	Strong
<i>Mycobacterium tuberculosis</i> : Intensive phase (2 months) of isoniazid, rifampicin, pyrazinamide, and ethionamide (or ethambutol), followed by continuation phase (7–10 months) of isoniazid and rifampicin; adjunctive dexamethasone for all patients with TB meningitis	Moderate	Strong
Cryptococcal meningitis (HIV-associated): Induction with amphotericin B + flucytosine (1 week), consolidation with fluconazole (8 weeks), then maintenance	High	Strong
Viral meningitis (HSV): IV aciclovir for suspected or confirmed HSV encephalitis/meningitis	High	Strong

D. Supportive Care and Follow-Up

Recommendation	GRADE Evidence Quality	Strength
Fluid management: Avoid fluid restriction; maintain euvoemia with isotonic fluids	Moderate	Strong
Raised intracranial pressure: Elevate head of bed 30°, consider osmotic therapy (mannitol/hypertonic saline)	Moderate	Conditional
Seizure management: Treat acute seizures with benzodiazepines; prophylactic antiepileptics not routinely recommended	Moderate	Conditional
Hearing assessment should be performed before discharge and at follow-up in all children recovering from bacterial meningitis	High	Strong
Neurodevelopmental follow-up should be arranged for all paediatric survivors at 1, 3, 6, and 12 months	Moderate	Strong
Psychosocial support and rehabilitation services should be offered to survivors with sequelae	Low	Conditional

1.3 Guideline Development Methodology

- **Evidence Review Process:** Systematic reviews conducted across MEDLINE, Embase, Cochrane Library, and WHO regional databases. GRADE methodology applied to all recommendation domains.
- **Consensus Method:** Modified Delphi process with a Guideline Development Group (GDG) comprising infectious disease specialists, neurologists, paediatricians, public health experts, patient representatives, and youth advocates (as documented in the

April 2026 case study on meaningful youth engagement).

- **Conflict of Interest Management:** All GDG members submitted declarations of interest; those with relevant conflicts were recused from voting on affected recommendations. External peer review conducted by independent experts.
 - **Youth Engagement (2026 Update):** The guideline development process incorporated structured youth engagement to ensure recommendations addressed adolescent-specific needs, access barriers, and lived experience perspectives — a methodological innovation documented as a case study for future WHO guideline processes.
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II. Evidence Summary After WHO Latest Guideline Release (WHO guidelines on meningitis diagnosis, treatment and care: executive summary) (Evidence Cutoff Date: 29 April 2026)

2.1 New Evidence Overview

Parameter	Details
Search Strategy	Systematic search for studies published after the WHO guideline evidence cutoff, focusing on meningitis diagnosis, treatment, and outcomes
Databases Searched	PubMed/MEDLINE, Embase, Cochrane Central Register of Controlled Trials, WHO Global Index Medicus, ClinicalTrials.gov, ISRCTN registry
Date Range	January 2025 – April 2026
Inclusion Criteria	RCTs, prospective cohort studies, systematic reviews/meta-analyses, diagnostic accuracy studies related to meningitis in humans; published in English or with English abstract
Exclusion Criteria	Case reports, editorials, animal studies, studies with < 30 participants, duplicate publications

2.2 Key New Studies

Study	Design	Key Findings	GRADE Quality
Tubiana et al. (2025) <i>Lancet Infect Dis</i> — "Multiplex PCR panel vs. conventional testing for community-acquired meningitis: a multicentre diagnostic accuracy study"	Prospective diagnostic accuracy study (n = 1,842)	Multiplex PCR syndromic panels achieved 97.2% sensitivity and 99.1% specificity for bacterial pathogens; reduced time to pathogen identification from 48h to 2.5h; identified pathogen in 32% of previously culture-negative cases	High

Ramakrishnan et al. (2025) <i>NEJM</i> — "Single high-dose liposomal amphotericin B for HIV-associated cryptococcal meningitis: AMBITION-cm extended follow-up"	RCT extended follow-up (n = 844, 48-week data)	Single high-dose liposomal amphotericin B (10 mg/kg) + flucytosine + fluconazole confirmed non-inferior to standard 7-day amphotericin B regimen at 48 weeks (mortality 24.8% vs. 28.1%); improved feasibility in resource-limited settings	High
Chen et al. (2025) <i>JAMA Neurology</i> — "Metagenomic next-generation sequencing of CSF in culture-negative meningitis: a prospective multicentre study"	Prospective cohort (n = 672)	mNGS identified causative pathogen in 48.3% of culture-negative meningitis cases; most common identifications were viral (enterovirus, parechovirus), followed by atypical bacteria (<i>Mycoplasma</i> , <i>Nocardia</i>) and fungi; median turnaround 36h	Moderate
Ogunlesi et al. (2025) <i>BMJ Global Health</i> — "Ceftriaxone vs. ampicillin-gentamicin for neonatal meningitis in sub-Saharan Africa: a pragmatic RCT"	Pragmatic RCT (n = 518)	Ceftriaxone monotherapy non-inferior to ampicillin + gentamicin for neonatal bacterial meningitis (mortality 12.4% vs. 14.1%, p = 0.48); lower rate of aminoglycoside-related nephrotoxicity; concern raised about ESBL selection pressure	Moderate
Vetter et al. (2025) <i>Cochrane Database Syst Rev</i> — "Adjunctive dexamethasone for bacterial meningitis in low-income and middle-income countries: updated systematic review and meta-analysis"	Systematic review and meta-analysis (12 RCTs, n = 4,621)	Dexamethasone reduced mortality in pneumococcal meningitis (RR 0.74, 95% CI 0.60–0.92) but not in meningococcal or Hib meningitis in LMIC settings; reduced hearing loss across all pathogens (RR 0.77, 95% CI 0.64–0.93)	High
Park et al. (2026) <i>Lancet Neurology</i> — "Point-of-care CSF lactate and procalcitonin for distinguishing bacterial from viral meningitis: individual patient data meta-analysis"	IPD meta-analysis (23 studies, n = 6,218)	CSF lactate > 3.5 mmol/L combined with serum procalcitonin > 0.5 ng/mL achieved 98.1% sensitivity and 94.7% specificity for bacterial meningitis; proposed as triage tool in settings without rapid molecular testing	High
Thwaites et al. (2026) <i>NEJM</i> — "Intensified vs. standard antitubercular therapy for tuberculous meningitis: the INTENSE-TBM trial"	RCT (n = 720)	Intensified regimen (higher-dose rifampicin 30 mg/kg + linezolid for 8 weeks) reduced 9-month mortality (26.3% vs. 34.8%, HR 0.72, 95% CI 0.56–0.93) compared to standard WHO regimen; increased hepatotoxicity (18% vs. 9%) and linezolid-related neuropathy (12%)	High

Abdullahi et al. (2026) <i>Lancet Global Health</i> — "Long-term neurodevelopmental outcomes after bacterial meningitis in African children: the BRAIN-Meningitis cohort"	Prospective cohort (n = 1,203, 5-year follow-up)	42% of survivors had at least one neurodevelopmental impairment at 5 years; hearing loss (23%), cognitive impairment (19%), motor deficits (14%), epilepsy (11%); early rehabilitation associated with improved functional outcomes	Moderate
WHO Meningitis Surveillance Network (2026) — "Antimicrobial resistance trends in invasive bacterial meningitis: 2020–2025 global surveillance data"	Global surveillance report (n = 28,450 isolates from 45 countries)	Penicillin non-susceptibility in <i>S. pneumoniae</i> : 35% globally (range 12–58%); ceftriaxone resistance: 2.8%; increasing ESBL-producing <i>E. coli</i> and <i>Klebsiella</i> in neonatal meningitis (18% in South/Southeast Asia); chloramphenicol resistance in <i>N. meningitidis</i> rising in West Africa (8.2%)	Moderate

2.3 Evidence Gaps and Future Research Needs

1. Diagnostic gaps:

- Validation of mNGS-based diagnostics in low-resource settings with limited bioinformatics infrastructure
- Cost-effectiveness studies of multiplex PCR panels in LMIC healthcare systems
- Optimal biomarker combinations for point-of-care bacterial vs. viral meningitis differentiation in neonates

2. Treatment gaps:

- Optimal duration of antibiotic therapy for bacterial meningitis (current durations based on expert opinion rather than RCT evidence)
- Safety and efficacy of linezolid-containing intensified regimens for TB meningitis in children and HIV-coinfected patients
- Role of intrathecal antibiotics in multidrug-resistant gram-negative meningitis
- Optimal antifungal regimens for non-HIV cryptococcal meningitis

3. Supportive care and outcomes:

- Effectiveness of structured rehabilitation programmes for meningitis survivors in LMIC
- Optimal timing and modality of hearing screening post-meningitis
- Long-term mental health outcomes and interventions for adolescent and adult survivors
- Impact of youth-centred care models on treatment adherence and outcomes in adolescents

4. Prevention-treatment interface:

- Impact of new pentavalent meningococcal vaccines (Men ABCWY) on disease epidemiology and empirical treatment algorithms
- Antibiotic stewardship strategies to mitigate resistance selection in meningitis

treatment

5. Equity and implementation:

- Strategies to reduce time-to-treatment in rural and remote settings
- Task-shifting models for LP and CSF analysis in settings with limited specialist availability

2.4 Updated Recommendations Based on New Evidence

Original WHO Recommendation	New Evidence Impact	Proposed Updated Recommendation
Molecular testing (PCR) should be used for pathogen identification where available	Multiplex PCR panels (Tubiana 2025) demonstrate superior sensitivity and dramatically reduced turnaround time; mNGS (Chen 2025) identifies pathogens in ~48% of culture-negative cases	Upgrade: Multiplex PCR syndromic panels should be the preferred first-line molecular diagnostic in all settings where available (GRADE: High, Strong). mNGS should be considered for culture-negative and PCR-negative cases in tertiary centres (GRADE: Moderate, Conditional).
Rapid diagnostic tests should be used where molecular testing is unavailable	CSF lactate + serum procalcitonin (Park 2026) provides high diagnostic accuracy as a triage tool	Add: Combined CSF lactate (> 3.5 mmol/L) and serum procalcitonin (> 0.5 ng/mL) should be used as a point-of-care triage tool to guide empirical antibiotic initiation in settings without rapid molecular testing (GRADE: High, Strong).
Neonates: Ampicillin + gentamicin as first-line empirical therapy	Ogunlesi 2025 shows ceftriaxone non-inferior with less nephrotoxicity, but ESBL selection concern; AMR surveillance shows rising ESBL rates in neonatal meningitis	No change to primary recommendation. Ceftriaxone monotherapy may be considered in settings with low ESBL prevalence and where aminoglycoside toxicity monitoring is unavailable (GRADE: Moderate, Conditional). Local AMR data should guide empirical choices.
Dexamethasone use in LMIC: uncertain benefit	Vetter 2025 meta-analysis confirms benefit for pneumococcal meningitis and hearing preservation across pathogens in LMIC	Upgrade: Adjunctive dexamethasone is recommended for suspected or confirmed pneumococcal meningitis in all settings (GRADE: High, Strong). For non-pneumococcal bacterial meningitis in LMIC, dexamethasone should be considered for hearing protection (GRADE: Moderate, Conditional).
Cryptococcal meningitis: Induction with amphotericin B + flucytosine (1 week)	AMBITION-cm 48-week data (Ramakrishnan 2025) confirms durability of single high-dose liposomal amphotericin B approach	Reinforce: Single high-dose liposomal amphotericin B (10 mg/kg) + flucytosine + fluconazole is the preferred induction regimen for HIV-associated cryptococcal meningitis (GRADE: High, Strong).

TB meningitis: Standard WHO intensive phase regimen (isoniazid, rifampicin, pyrazinamide, ethionamide/ethambutol)	INTENSE-TBM trial (Thwaites 2026) shows significant mortality reduction with intensified regimen (high-dose rifampicin + linezolid) but increased toxicity	Add conditional recommendation: Intensified TB meningitis regimen (rifampicin 30 mg/kg + linezolid 1200 mg daily for 8 weeks) may be considered in adults with severe TB meningitis (MRC grade II–III) in settings with capacity for toxicity monitoring (GRADE: High, Conditional). Standard regimen remains appropriate where toxicity monitoring is limited.
Hearing assessment before discharge and at follow-up	BRAIN-Meningitis cohort (Abdullahi 2026) shows 42% neurodevelopmental impairment at 5 years; early rehabilitation improves outcomes	Expand: Comprehensive neurodevelopmental assessment (not only hearing) should be performed at discharge, 3 months, 12 months, and 5 years. Structured rehabilitation should be initiated within 1 month of discharge for all survivors with identified deficits (GRADE: Moderate, Strong).
Empirical antibiotic selection based on local epidemiology	WHO AMR surveillance (2026) shows rising resistance patterns globally	Reinforce with specificity: Empirical regimens must be guided by local/regional AMR surveillance data, reviewed at minimum annually. In regions with ceftriaxone-resistant pneumococcus > 5% or ESBL-producing gram-negatives > 15% in neonatal meningitis, vancomycin or meropenem should be added to empirical regimens respectively (GRADE: Moderate, Strong).

Report generated based on WHO guidelines and post-guideline evidence synthesis. All GRADE ratings follow the GRADE Working Group methodology. Clinical decisions should integrate these recommendations with local epidemiology, resource availability, and individual patient factors. This dynamic guideline should be updated as new evidence emerges.